

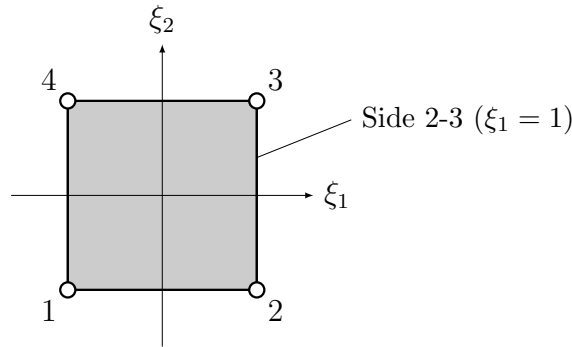
Problem set 7

Problem 1

Consider the shape functions for the AMC element:

$${}^a\mathbf{h}_i(\boldsymbol{\xi}) = \begin{bmatrix} (1 + \xi_1^i \xi_1)(1 + \xi_2^i \xi_2)(2 + \xi_1^i \xi_1 + \xi_2^i \xi_2 - \xi_1^2 - \xi_2^2)/8 \\ b(1 + \xi_1^i \xi_1)(\xi_2^i + \xi_2)(\xi_2^2 - 1)/8 \\ -a(\xi_1^i + \xi_1)(\xi_1^2 - 1)(1 + \xi_2^i \xi_2)/8 \end{bmatrix}^T$$

where $\boldsymbol{\xi}^i = \{\xi_1^i, \xi_2^i\}^T$ are the local coordinates of node $i = 1, \dots, 4$.



a) Show that, for $i, j = 1, \dots, 4$, the Kronecker delta property holds:

$${}^a\mathbf{h}_i(\boldsymbol{\xi}^j) = \begin{bmatrix} \delta_{ij} \\ 0 \\ 0 \end{bmatrix}^T$$

where δ_{ij} is the Kronecker symbol.

b) Show that the generalized displacement function defined by

$${}^e u_3^h(\boldsymbol{\xi}, t) = \sum_{i=1}^4 {}^a\mathbf{h}_i(\boldsymbol{\xi}) \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \end{bmatrix} = {}^a\mathbf{H}(\boldsymbol{\xi}) {}^e \mathbf{q}(t)$$

is a non-conforming one by evaluating u_3^h , θ_1^h and θ_2^h on the side 2-3 ($\xi_1 = 1$) and by showing that θ_2^h could be discontinuous along this edge.

Problem 2

Demonstrate that, in the AMC plate bending element, selecting a displacement function that includes the quartic terms x^4 and y^4 instead of the mixed cubic terms x^3y and xy^3 results in displacement discontinuities along the element boundaries.

Problem 3

Consider the following shape functions:

$${}^a\mathbf{h}_i(\boldsymbol{\xi}) = \begin{bmatrix} f_i(\xi_1)f_i(\xi_2) \\ bf_i(\xi_1)g_i(\xi_2) \\ -ag_i(\xi_1)f_i(\xi_2) \end{bmatrix}^T$$

where $\boldsymbol{\xi}^i = \{\xi_1^i, \xi_2^i\}^T$ are the local coordinates of node $i = 1, \dots, 4$ and f_i and g_i are the Hermite cubic shape functions:

$$f_i(\xi) = (-\xi^i\xi^3 + 3\xi^i\xi + 2)/4, \quad g_i(\xi) = (\xi^3 + \xi^i\xi^2 - \xi - \xi^i)/4.$$

Show that the generalized displacement function, defined by

$${}^e u_3^h(\boldsymbol{\xi}, t) = \sum_{i=1}^4 {}^a\mathbf{h}_i(\boldsymbol{\xi}) \begin{bmatrix} {}^e d^i(t) \\ {}^e \theta_1^i(t) \\ {}^e \theta_2^i(t) \end{bmatrix} = {}^a\mathbf{H}(\boldsymbol{\xi}) {}^e \mathbf{q}(t),$$

has twist $\frac{\partial^2}{\partial \xi_1 \partial \xi_2} {}^e u_3^h$ zero at the four nodal points.